ISE 3434: Applied Optimization

Assignment #1

Due Date: September 15, 2026, 11:59pm

Directions:

- Submit one PDF file (your write-up) and one Jupyter notebook (.ipynb) file to Canvas by the due date above. Late submissions are accepted according to the late policy in the syllabus.
- For each problem, your PDF should clearly define the model (variables, objective, constraints, and nonnegativity & integer restrictions), and briefly describe your solution along with the optimal objective value.
- Your notebook should contain working code that prints your results.
- Grading considers both the correctness and clarity of your written solutions, and the correctness and readability of your code.
- Before starting, make sure Gurobi is installed with an academic license and that the gurobipy package is working in your Python.

Problem 1: Green Energy Investments (25 points)

EcoPower is deciding where to invest in renewable energy projects: solar farms, wind turbines, or geothermal plants. Each project type can be built in one of three regions (North, Central, South). EcoPower can fund at most two projects overall, and no more than one project per region. The table below shows the net present value (NPV) and capital required for each project-region combination. The company has a total budget of $15 million and wants to maximize total NPV.

Project Type	Region	NPV (Million $)	Capital Required (Million $)
Solar Farm	North	4	5
Solar Farm	Central	3.5	4
Solar Farm	South	4.5	6
Wind Turbine	North	5.5	7
Wind Turbine	Central	4	5
Wind Turbine	South	5	6
Geothermal Plant	North	6	8
Geothermal Plant	Central	4.5	6
Geothermal Plant	South	6.5	9

Model the problem as a binary integer program and solve it using Gurobi.

To submit

- Model description: variables, objective, constraints, and nonnegativity & integer restrictions.
- Your code as part of the single .ipynb file, with results printed.
- A brief written summary of your solution, including the optimal objective value.

Problem 2: Airport Ground Crew Scheduling (25 points)

An airport needs to staff its ground crew across the week. Crew members work fixed three-day shift patterns, each with its own daily wage, and part-time workers can be hired for a single day at a higher rate to cover any remaining need. The goal is to meet each day's staffing requirement at the lowest total cost.

Shift	Days Worked	Daily Wage
Shift 1	Mon, Tue, Wed	$600
Shift 2	Tue, Wed, Thu	$550
Shift 3	Wed, Thu, Fri	$650
Shift 4	Thu, Fri, Sat	$620
Shift 5	Fri, Sat, Sun	$580
Shift 6	Mon, Thu, Sat	$700
Shift 7	Tue, Fri, Sun	$680

Part-time workers can be hired at a rate of $800 per day.

Day	Crew Members Required
Monday	150
Tuesday	180
Wednesday	200
Thursday	220
Friday	250
Saturday	300
Sunday	100

Model the problem as an integer program and solve it using Gurobi.

To submit

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- Your code as part of the single .ipynb file, with results printed.
- A brief written summary of your solution, including the optimal objective value.

Problem 3: Storage Optimization for Moving Boxes (25 points)

You are helping a friend move. They have 15 items of different volumes (in cubic feet, listed below) that need to be packed into moving boxes. They have 8 boxes available, each with a 50 cubic foot capacity, and the moving company charges $30 per box used. Your friend will pay you $30 for every box you manage to save by packing efficiently. The goal is to minimize the number of boxes used without exceeding any box's capacity.

Item	Volume	Item	Volume	Item	Volume
1	12	6	8	11	11
2	7	7	15	12	18
3	19	8	22	13	16
4	10	9	9	14	13
5	24	10	14	15	17

Model the problem as a binary integer program and solve it using Gurobi.

To submit

- Model description: variables, objective, constraints, and nonnegativity & integer restrictions.
- Your code as part of the single .ipynb file, with results printed.
- A brief written summary of your solution, including the optimal objective value.

Problem 4: Festival Security Staffing (25 points)

A town festival needs security coverage overnight, split into 8 work shifts. Because some shifts overlap in time, a single guard cannot be assigned to two shifts that conflict with one another. The table below lists each shift along with the other shifts it conflicts with (the relationship goes both ways). Every shift must be covered by exactly one guard, and the organizers want to hire as few guards as possible.

Shift	Conflicts With
1	2, 3
2	1, 3, 4
3	1, 2, 5
4	2, 5, 6
5	3, 4, 6, 7
6	4, 5, 7
7	5, 6, 8
8	7

Model the problem as a binary integer program and solve it using Gurobi.

To submit

- Model description: variables, objective, constraints, and nonnegativity & integer restrictions.
- Your code as part of the single .ipynb file, with results printed.
- A brief written summary of your solution, including the optimal objective value.